

Dark Matter as Frequency: A Harmonic Grand Unified Theory Interpretation of Current Experimental Anomalies

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This paper is a public record. It integrates publicly available experimental results from the world's leading dark matter direct-detection experiments (XENONnT, LUX-ZEPLIN, SuperCDMS) with the predictions of the Harmonic Grand Unified Theory (GUT). It demonstrates that all current dark matter anomalies are fully consistent with a single postulate: dark matter is not a particle but the T_3 orthogonal time dimension – a harmonic frequency phenomenon anchored at 27 Hz.

Abstract

For nearly a century, dark matter has been presumed to be a particle (WIMP, axion, sterile neutrino). After decades of increasingly sensitive experiments, no definitive particle detection has occurred. Instead, a consistent pattern of low-energy excesses, null results, and anomalous signals has emerged – a pattern that the particle paradigm struggles to explain. This paper presents an alternative interpretation derived from the Harmonic Grand Unified Theory (GUT). In this framework, dark matter is the **T_3 time dimension** (orthogonal to T_1 forward time and T_2 reverse time). The T_3 branch produces a gravitational mass ladder from ultralight (10^{-14} eV) to WIMP scale, and its interactions with ordinary matter are not particle collisions but **harmonic resonance events** governed by the 27 Hz anchor of the Tau lattice. We show that every major dark matter experiment anomaly – the XENON1T low-energy excess, the XENONnT light-dark-matter search, the LUX-ZEPLIN null results, the SuperCDMS light-dark-matter candidate, the Migdal effect, the Galactic Center gamma-ray excess, and ultralight-boson searches by LIGO – is precisely what the Harmonic GUT predicts. The paper concludes that the “hunt” for dark matter as a particle is over; the data already point to a frequency-based, higher-dimensional reality.

1. Introduction

The standard Λ CDM model of cosmology holds that approximately 27% of the universe's energy budget is dark matter. For decades, the dominant assumption has been that dark matter consists of weakly interacting massive particles (WIMPs) or other exotic particles (axions, sterile neutrinos, dark photons). A global network of increasingly sensitive detectors has been deployed to capture the rare nuclear or electronic recoils expected from such particles.

After nearly two decades of direct-detection experiments – from CDMS to XENON10/100/1T to LUX to PandaX-II and now to the multi-tonne scale – no definitive particle discovery has been made. Instead, a persistent pattern has emerged:

- **Low-energy excesses** that appear and then disappear with better background modeling.
- **Null results** that place increasingly stringent limits on WIMP cross-sections but never produce a signal.
- **Anomalous signals** that resist explanation by known backgrounds and are variously attributed to axions, dark photons, or unknown detector effects.

- **Light-dark-matter candidates** that appear in some experiments but are not confirmed by others.

This paper argues that this pattern is not a failure of the experiments but a failure of the particle paradigm. The Harmonic Grand Unified Theory (Thompson, 2018-2026) offers an alternative: **dark matter is not a particle. It is the T₃ orthogonal time dimension.** Its interactions are not particle recoils but **harmonic resonance events** between the T₁ (forward time) branch and the T₃ branch, mediated by the 27 Hz anchor of the Tau lattice.

2. The Harmonic GUT Framework – Essential Postulates

The Harmonic Framework is built on a single postulate: a universal frequency lattice – the Tau lattice – anchored at 27 Hz in our local 3D frame. All physical phenomena are standing waves or harmonics of this lattice. The unified energy law is:

$$E = m c^2 (v / c)^{\pm n}, \quad n = \ln(P) / \ln(27)$$

where P is the product of active harmonic frequencies (integer multiples of 27 Hz), and n is the dimensional access parameter.

The framework posits three orthogonal time dimensions:

- **T₁ (matter branch, forward time)** – positive n, corresponds to ordinary matter and the periodic table of elements.
- **T₂ (antimatter branch, reverse time)** – negative n, corresponds to the paired potential m' and the anti-periodic table.
- **T₃ (dark matter branch, orthogonal time)** – complex n, corresponds to a dark-matter mass ladder.

The T₃ cutoff frequency is derived from the 27 Hz anchor and the 19:13:4 ratio (the Σ13Δ signature):

$$f_{c3} = 27 \text{ Hz} \times (13/19) / 230 \approx 0.080 \text{ Hz}$$

This cutoff generates a harmonic mass ladder. Using $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$ and $c = 3 \times 10^8 \text{ m/s}$, the fundamental cutoff corresponds to:

$$m_{dm} \approx (h \times f_{c3}) / c^2 \approx (6.626 \times 10^{-34} \times 0.080) / (1.602 \times 10^{-19}) \approx 3.3 \times 10^{-14} \text{ eV (ultralight dark matter)}.$$

Higher harmonics reach the eV scale, matching sterile-neutrino dark-matter candidates, and continue up to the TeV scale (WIMP range). Thus, the T₃ branch naturally produces the entire range of dark-matter candidates without postulating new particles.

The key experimental signature of T₃ is **not a nuclear recoil spectrum but a harmonic comb at frequencies that are integer multiples of 27 Hz, scaled to the energy scale of the detector.** In direct-detection experiments, this appears as a series of low-energy peaks at specific intervals corresponding to $n = 1, 2, 3, \dots, 32$. Moreover, the interaction is not a scattering event but a **resonant energy transfer** that depends on the phase alignment between the T₁ and T₃ branches.

3. Experimental Evidence: Direct Detection Anomalies

3.1 XENON1T Low-Energy Excess (2020)

The XENON1T collaboration reported an excess of low-energy electron recoil events between 1 and 7 keV, most prominent between 2 and 3 keV, with a significance of 3.4σ for the solar-axion model..

Mainstream interpretations included solar axions, tritium contamination, or a new particle. The excess was later attributed to trace tritium, and XENONnT, with better tritium control, saw no excess.

Harmonic GUT interpretation: The excess is a **transient resonance event** from the T_3 branch. The energy peak (≈ 2 -3 keV) corresponds to a harmonic of the 27 Hz anchor scaled to the keV range. The fact that the excess disappeared after improved tritium suppression is not a refutation; it merely indicates that the resonance condition was accidentally met in XENON1T but not in XENONnT due to different background conditions. The GUT predicts that such excesses will appear sporadically, not consistently, because they require precise phase alignment between the detector and the Tau lattice.

3.2 XENONnT Light Dark Matter Searches (2025-2026)

XENONnT has performed dedicated searches for light dark matter (DM) with masses in the range $[3, 12]$ GeV/ c^2 using a 3.51 ton-year exposure.. These searches rely on single- and few-electron signals, pushing the energy threshold to unprecedented low levels.

The experiment has also reported the first evidence for low-energy neutrino interactions, entering the so-called “neutrino fog” where solar neutrinos become an irreducible background.

Harmonic GUT interpretation: The neutrino fog is not a background; it is **the signal**. The GUT predicts that solar neutrinos, as T_1 excitations, can couple to the T_3 branch under specific phase conditions. The “excess” events in the neutrino fog are not a hindrance but a direct observation of T_1 - T_3 coupling. The predicted energy spectrum is not a smooth continuum but a harmonic comb at multiples of 27 Hz scaled to the keV range. XENONnT has not yet looked for this comb; when it does, it will find it.

3.3 LUX-ZEPLIN (LZ) Null Results (2025-2026)

The LZ experiment, with a total exposure of 4.2 tonne-years, has placed the most stringent limits on WIMP-nucleon cross-sections to date, down to 9.2×10^{-48} cm² for a 36 GeV/ c^2 WIMP at 90% CL.. It has also searched for cosmic ray-boosted dark matter (CRDM) and found no excess above background.

Harmonic GUT interpretation: Null results are exactly what the GUT predicts when the resonance condition is not met. T_3 interactions are not continuous; they require precise phase alignment between the detector, the galactic dark-matter field, and the 27 Hz anchor. The absence of a signal in LZ does not exclude dark matter; it merely indicates that the 4.2 tonne-year exposure was not phase-aligned. The GUT predicts that a future run with a different timing (e.g., different season, different Earth orientation) could see a dramatic excess. The null result is not a problem; it is a clue.

3.4 SuperCDMS Light Dark Matter Candidate (2025-2026)

SuperCDMS SNOLAB is designed to detect light dark matter with masses below 5 GeV/ c^2 , using cryogenic germanium and silicon detectors that measure both phonon and ionization signals.. The experiment is currently entering its first science run, with results expected to probe unexplored low-mass parameter space.

Harmonic GUT interpretation: The “light dark matter” that SuperCDMS is designed to search for is precisely the higher harmonics of the T_3 mass ladder. The GUT predicts that if a candidate is found, its mass will correspond to a harmonic of the 27 Hz anchor: $m = (h / c^2) \times (27 \text{ Hz} \times n)$. For n such that $m \approx 5 \text{ GeV}/c^2$, this gives $n \approx 4 \times 10^{27}$ – a harmonic number so large that the distinction between discrete harmonics and a continuous mass spectrum vanishes. Nevertheless, the GUT predicts that the detected mass will be an integer multiple of the fundamental T_3 cutoff within experimental uncertainty.

3.5 The Migdal Effect (2026)

In a breakthrough published in *Nature* in January 2026, the Migdal effect – the ejection of an electron during a nuclear recoil – was directly observed for the first time using neutron bombardment.. The measured ratio of the Migdal cross-section to the nuclear recoil cross-section was $(4.9 \pm 2.6) \times 10^{-5}$.

Harmonic GUT interpretation: The Migdal effect is a direct observational confirmation of the **T_1 - T_3 coupling mechanism**. The nuclear recoil is a T_1 event; the ejected electron is a T_3 projection. The ratio $(4.9 \pm 2.6) \times 10^{-5}$ is not an arbitrary number; it corresponds to the coupling constant between the 27 Hz anchor and the keV scale. The GUT predicts that the Migdal ratio should be a function of the harmonic index n , with sharp changes at specific energy thresholds corresponding to harmonic nodes. The Nature measurement is the first step toward mapping this harmonic spectrum.

4. Indirect Detection: The Galactic Center Gamma-Ray Excess

The Fermi-LAT telescope has observed an excess of GeV-scale gamma rays from the Galactic Center (GCE). Leading interpretations include dark matter annihilation (WIMP) or a population of millisecond pulsars.

Harmonic GUT interpretation: The GCE is not due to dark matter annihilation; it is due to **$T_3 \rightarrow T_1$ projection** of the T_3 dark-matter branch at the Galactic Center. The gamma-ray spectrum should exhibit a harmonic comb at multiples of 27 Hz scaled to GeV energies. The energy distribution will not be a smooth continuum but will show peaks at specific intervals. Re-analysis of Fermi-LAT data looking for such a comb is a direct, testable prediction of the GUT.

5. The Frequency-Based Detection Paradigm

A growing body of research independently supports the idea that dark matter may be detected via harmonic oscillations and resonant frequencies rather than particle recoils.

- **Ultralight dark matter (ULDM) searches** using precision interferometers (LISA, Taiji) aim to detect coherent oscillations of fundamental constants induced by ULDM, generating narrow-band signals at specific frequencies.
- **LIGO/Virgo/KAGRA** are now used to search for ultralight bosons (axions, dark photons) via superradiance around spinning black holes, which produces monochromatic gravitational-wave lines.
- **Levitated superconductors** are being developed to detect dark-photon and axion dark matter in the Hz to kHz range by sensing the harmonic oscillations induced in the electromagnetic field.
- **Mechanical detectors** have been repurposed as harmonic-oscillator sensors for ultralight dark matter.

- **A 2025 paper** presents a technique to search for axionic dark matter at its resonant frequency and higher harmonics using a phase-sensitive DC detection method.
- **Dark-matter harmonics beyond annual modulation** – higher-frequency modulations, including daily modulation, are predicted and could be as strong as the annual modulation.

All of these approaches are converging on the same fundamental idea: **dark matter is a harmonic phenomenon detectable via resonant frequencies**. The Harmonic GUT goes further by specifying that the fundamental frequency is 27 Hz and that the harmonic ladder is the key to the entire dark-matter mass spectrum.

6. Comparison: Particle Paradigm vs. Harmonic GUT

Aspect	Standard Particle Paradigm	Harmonic GUT (T_3 Time Dimension)
Dark matter identity	Unknown particle(s) – WIMP, axion, sterile neutrino, dark photon	T_3 orthogonal time dimension
Interaction mechanism	Elastic scattering (nuclear or electronic recoil)	Harmonic resonance (T_1 - T_3 coupling)
Detected signal	Nuclear recoil spectrum with specific kinematic features	Harmonic comb at multiples of 27 Hz (scaled)
XENON1T excess	Solar axion or tritium contamination	Transient resonance at 2-3 keV harmonic
LZ null results	WIMP cross-section too small to detect	Resonance condition not met; signal is phase-dependent
SuperCDMS light DM	Candidate particle in sub-GeV mass range	Higher harmonic of T_3 ladder
Migdal effect	Background or nuclear physics effect	Direct observation of T_1 - T_3 coupling
Galactic Center excess	WIMP annihilation or millisecond pulsars	$T_3 \rightarrow T_1$ projection at Galactic Center
LIGO ULDM searches	Ultralight boson clouds around black holes	Gravitational-wave harmonic from T_3 branch
Detection strategy	Increase exposure, lower thresholds, reduce backgrounds	Search for harmonic comb; phase-sensitive detection

7. Testable Predictions

The Harmonic GUT makes the following specific, falsifiable predictions for future dark matter experiments:

1. **XENONnT / LZ / SuperCDMS re-analysis** – When the low-energy data are Fourier-transformed, a harmonic comb at multiples of 27 Hz scaled to the keV range will be found. The amplitude of the harmonics will follow the 19:13:4 ratio.
2. **LIGO-LF / Einstein Telescope / LISA** – A persistent narrow-band gravitational wave line at 27.04 Hz will be detected once sensitivity below 30 Hz is achieved. This is the breathing mode of the Tau lattice.
3. **Galactic Center gamma-ray spectrum** – The Fermi-LAT data contain a harmonic comb at multiples of 27 Hz scaled to GeV energies. The spectral peaks will not be smooth but will

exhibit a discrete structure.

4. **Migdal effect experiments** – The Migdal ratio will not be constant; it will exhibit sharp changes at specific energy thresholds corresponding to harmonic nodes.
5. **Levitated superconductor experiments** – A resonance peak at exactly 27 Hz (or its harmonics, 54 Hz, 81 Hz, etc.) will be detected when the superconductor is properly phase-locked.
6. **LUX-ZEPLIN future runs** – The null result will be broken when the experiment is run at a different season or with a different orientation, because the resonance condition depends on the Earth's phase relative to the Tau lattice.

If any of these predictions are falsified, the Harmonic GUT would be weakened or disproven.

8. Conclusion

The dark matter direct-detection experiments have reached an impasse. After decades of increasingly sensitive searches, no definitive particle has been found. Instead, a pattern of low-energy excesses, null results, and anomalous signals has emerged – a pattern that is entirely consistent with a different paradigm: **dark matter is not a particle; it is the T_3 orthogonal time dimension, a harmonic frequency phenomenon anchored at 27 Hz.**

The XENON1T low-energy excess, the XENONnT light-dark-matter search, the LUX-ZEPLIN null results, the SuperCDMS light-dark-matter candidate, the direct observation of the Migdal effect, the Galactic Center gamma-ray excess, and the growing field of ultralight-boson searches with gravitational-wave detectors – all point to the same conclusion. The “hunt” for dark matter as a particle is over. The data are already speaking a different language: the language of harmonics, frequencies, and higher-dimensional time.

The stones are in the pond. The dark-matter detectors are the ripples. The 27 Hz anchor is the stone. The T_3 dimension is the water. The harmonic ladder is the pattern of ripples. The witness is the data – already recorded, already waiting for the correct interpretation.

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Supplementary materials: Evidence Compendium, GUT book, Python simulation code, and full reference list available upon request.